Vansh

vansh.s.works@gmail.com | LinkedIn | GitHub | LeetCode | +91 7742656072

Education

B.Tech in Computer Science Engineering, Vellore Institute of Technology *Sept 2023 – Aug 2027* | ***8.81 CGPA***
Class XII, Floradale School, Ganganagar, Rajasthan ***84.8%***
Class X, D.A.V. Public School, Suratgarh, Rajasthan ***95.33%***

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript
Web Technologies & Libraries: React, Node.js, Express.js, Next.js, Flask, FastAPI, HTML/CSS, Tailwind CSS, Shadcn, Socket.io, JWT
Machine Learning & AI: TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Large Language Models (LLMs), Prompt Engineering
Databases: NoSQL, MongoDB, MySQL, PostgreSQL, SupaBase, Redis (Upstash), Oracle Database, Prisma
Cloud & DevOps: AWS, Oracle Cloud Infrastructure (OCI), Docker, CI/CD
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs

Experience

Web Development Intern — KL LAB *May 2025 – Jul 2025 | Remote*

- Engineered secure and scalable **REST APIs** using **Node.js** and **Express.js**, implementing **JWT-based authentication** that protected **20+ backend endpoints** and reduced unauthorized access attempts by **30%**.
- Modeled **MongoDB schemas** via **Mongoose**, accelerating data retrieval efficiency for user, blog, and comment workflows by **40%** and reducing query latency to **under 50ms**.
- Deployed backend systems to **cloud infrastructure**, collaborating with a 4-person frontend team to achieve seamless **full-stack integration** and maintain **99.9% uptime** post-launch.

Technical Team Core Member — AWS Club, VIT Bhopal *Oct 2024 – Jul 2025 | Bhopal, MP*

- Orchestrated **5+ technical workshops** on **AWS fundamentals** for **100+ members**, driving a **20% increase** in club engagement and leading team-based **cloud architecture projects**.

Projects

AI-Powered ATS Resume Screener | **GitHub** | **Live Demo** ***Next.js, React, Shadcn, Redis, Prisma, PostgreSQL, TypeScript, LLMs***

- Architected an **enterprise ATS simulation platform** that evaluates resumes concurrently against **Legacy** (Taleo, Workday) and **Modern** (Greenhouse, Eightfold) systems, utilizing distinct parser strictness and **semantic evaluation rules**.
- Reduced API latency and token consumption by **50%** via **LLM prompt consolidation** and implemented **SHA-256 normalised response caching** via **Upstash Redis**, reducing repeated scan latency by **97%** — from **37 seconds to under 1 second**.
- Built a **high-availability inference pipeline** cascading from **Gemini 2.5 Flash** to **Groq** (Llama 3.3 70B - Llama 3.1 8B) with **exponential back-off** and a **three-tier sliding window rate limiter** (Redis - PostgreSQL - fail-open), guaranteeing near **100% uptime** during provider rate limits.
- Designed a highly concurrent data architecture featuring a fail-open **Prisma 7 rate-limiting system** and unified **PostgreSQL logging transactions**, deployed behind edge-protected **Clerk v6 middleware**.

Real-Time Network Intrusion Detection System | **GitHub** | **Live Demo** ***React, Express, Node.js, MongoDB, XGBoost, Docker***

- Developed a high-performance **XGBoost binary classifier** pipeline utilizing **Python, NumPy, and Pandas**, trained on **421,959 cleansed network flows** from the CIC-DDoS2019 dataset, achieving **99.85% accuracy** in distinguishing malicious from benign traffic.
- Secured an **Express backend** with **JWT auth**, rate limiting, Helmet security headers, Winston structured logging, and **bcryptjs password hashing**, enabling **sub-second IP blocklist enforcement** for detected threats via **3 controlled mitigation endpoints**.
- Programmed a **real-time SOC dashboard** with **React and Socket.io**, streaming live threat alerts, packet graphs, **geo-IP enrichment**, interactive world threat maps, and a **1-second ML prediction** button, with updates rendered in **under 2 seconds**.
- Containerized a 3-service architecture across **3 multi-stage Dockerfiles** to reduce production image sizes by **up to 90%**, and built a **GitHub Actions CI/CD workflow** to automate build validation and **Docker Hub** publishing – **Kubernetes-ready** with rolling updates and health probes.

SureBet | **GitHub** | **Live Demo** ***React, Node.js, Express.js, MongoDB, Socket.io***

- Pioneered a **real-time sports arbitrage** discovery platform scanning **15+ sportsbooks**, featuring instant data updates via **WebSockets** to deliver high-value betting opportunities **30% faster**.
- Formulated advanced **filtering and sorting algorithms** across **10+ data points**, enabling users to isolate high-yield opportunities **50% faster**.
- Constructed a **full-stack architecture** using **Node.js, Express, and MongoDB**, facilitating **scalable backend logic** and providing secure data handling for over **100+ data points** and improved data retrieval speeds **2x**.
- Implemented a **virtualized data table** within the dashboard with sortable columns and **live stake breakdowns**, utilizing **virtual scrolling** to render **5,000+ data rows** without frame drops, improving client-side load speeds by **60%**.

Certifications

- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- Oracle Cloud Database Services 2025 Certified Professional